

ON AN ITERATED BINARY-STRING REWRITING PROCESS

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ABSTRACT. We study a deterministic rewriting process F on finite binary strings that carries an integer counter s and halts precisely when s falls below 2; the central question is whether the trajectory generated from the seed $L_0 = 10$ is infinite. We reduce this question to a scalar inequality on the counter sequence (s_n) and analyse it from several directions. The halting value $s = 0$ is excluded unconditionally at every even index by an exact prefix invariant ($r_1(L_n) = 1 \iff n$ even), and at odd indices under the mild bound $s_{n-2} \geq 4$. We prove the two-step monotonicity $s_n \geq s_{n-2}$ at every even-parity step whose predecessor is also of even parity, via an even-step floor $s_n \geq 3\lfloor(s_{n-2} - 3)/2\rfloor$, and we give an exact recurrence for the 1-count $\nu(L_n)$ that fixes the parity sequence in closed form. A counter-inert tail lemma yields an exact decomposition of the even-step counter into a count term and an alignment-filtered size surplus, and shows that the count alone provably cannot close the residual gap. The map is recast as a faithful integer conjugate of Collatz type, as a gap-vector rewrite, and as a scalar process with no closed recurrence. Two independent from-scratch engines yield the counter through $s_{26} = 88\,613\,778$ and the parity through $\nu(L_{26}) = 222\,569\,758$, confirm two-step monotonicity throughout $3 \leq n \leq 26$, and localize the single remaining obstruction to even-parity steps with odd-parity predecessors. A dedicated section explains, with the forward and backward routes side by side, why the residual inequality resists elementary methods. The central theorem and the full monotonicity conjecture remain open.

CONTENTS

1. Introduction	2
Conventions	2
1.1. Summary of results	3
2. Elementary structure and the worked trajectory	3
3. Odd parity forces survival	4
4. The run-length calculus	4
5. Structural lemmas	4
6. Prefix dynamics and the all-even profile	5
7. The reduction to a scalar inequality	6
7.1. An improved even-step floor and a proven sub-case	7
8. The counter-inert tail and body localization	7
9. The ν -recurrence and the parity dynamics	8
10. Equivalent formulations	8
10.1. The pair-and-fill identity	8
10.2. The integer form	9
10.3. The gap-vector form	9
10.4. The scalar form	10
11. The backward viewpoint	10

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12. Computational results	11
12.1. The residual obstruction, localized	11
13. Why the problem resists	12
13.1. Two routes, one obstruction	12
13.2. Why that obstruction is a wall	12
13.3. What the evidence does and does not settle	13
14. Status	13
Appendix A. Algorithms	14
Appendix B. Computational evidence	14
Appendix C. Computational programs	15
Appendix D. Note on the use of artificial intelligence	15

1. INTRODUCTION

The process studied here is easy to define and stubbornly resistant to complete analysis. One iterates a fixed rewriting rule on binary strings; the rule carries a small integer counter and is engineered to stop the iteration the instant that counter drops below the threshold 2. From the seed $L_0 = 10$ the iteration appears to run forever, with the counter exploding to infinity after a single, isolated brush with the threshold. Proving that it truly never stops is the problem.

The architecture of the analysis is the following: modulo one structural fact—that a particular halting value never occurs on the orbit of 10—the whole question is equivalent to a single inequality about the size of the counter, driven by a transparent geometric mechanism. We establish that structural fact, prove the governing inequality in a structurally identified sub-case, push the computational frontier outward, and pin down sharply the one configuration on which a complete proof still founders.

Conventions. Strings are finite words over $\{0, 1\}$; positions are numbered $1, 2, 3, \dots$ from the left. For $w \in \{0, 1\}^*$ we write $\nu(w)$ for the number of 1's and $\omega(w)$ for the number of odd-length maximal runs of 1's. Concatenation is denoted by juxtaposition or a dot, and u^k is the k -fold concatenation of u . We call a string *even* (resp. *odd*) when ν is even (resp. odd); this “parity” refers to ν , not to the index n (Table 2).

Definition 1 (The map F). For $L \in \{0, 1\}^*$, the value $F(L)$, when defined, is computed in three steps.

Step 1 (pad; seed the counter). Let $k = \nu(L)$. If k is even, append 00 to L and set $s := 0$; if k is odd, append 0001 to L and set $s := -1$. Call the result $L^{(1)}$; note $\nu(L^{(1)})$ is even.

Step 2 (pair the 1's; fill and count). Let $p_1 < \dots < p_{2m}$ be the positions of the 1's in $L^{(1)}$, grouped into consecutive pairs $(p_1, p_2), \dots, (p_{2m-1}, p_{2m})$. For $j = 1, \dots, m$ in turn: set positions $p_{2j-1}, \dots, p_{2j} - 1$ to 1; set position p_{2j} to 0; and add $p_{2j} - p_{2j-1} - 1$ to s . Call the result $L^{(2)}$.

Step 3 (halt or extend). If $s < 2$, the process halts and $F(L)$ is undefined. Otherwise $F(L) := L^{(2)} \cdot (0110)^{s-2}$.

Informally, Step 2 takes the 1's two at a time, fills the gap between each pair into a solid block of 1's, deletes the block's right endpoint, and tallies the filled gap; the counter s is the total filling. Step 1 makes the count of 1's even so that the pairing is exhaustive, and Step 3 reads the total filling and, when there was enough, glues on a tail whose length grows with s .

Definition 2 (Trajectory). Set $L_0 := 10$ and $L_{n+1} := F(L_n)$ whenever the right-hand side is defined. Write s_n for the counter computed during $F(L_n)$.

Theorem 1 (Non-halting; the central claim, open). L_n is defined for every $n \geq 0$; equivalently, $s_n \geq 2$ for all n .

Theorem 1 is not proved here, and we believe it is not within reach of elementary methods. What we provide is a reduction to a single scalar inequality and a sharply delimited residual obstruction; §14 states the status with precision.

1.1. Summary of results. The main results are the following.

- (i) A prefix-alternation lemma (Lemma 10): $r_1(L_n) = 1 \iff n$ even, where r_1 is the length of the first maximal 1-run.
- (ii) Exclusion of the all-even halting profile $s_n = 0$ (Proposition 2): unconditionally at every even index and, at odd indices, whenever $s_{n-2} \geq 4$.
- (iii) An even-step floor and a proven sub-case of two-step monotonicity (Theorem 2): $s_n \geq 3\lfloor(s_{n-2} - 3)/2\rfloor$, whence $s_n \geq s_{n-2}$ once $s_{n-2} \geq 12$.
- (iv) An exact recurrence for $\nu(L_n)$ (Proposition 5) which closes the parity dynamics.
- (v) A counter-inert tail lemma (Lemma 11) and an exact body localization of the even-step counter (Observation 1): s_n is a functional of the body $L_{n-1}^{(2)}$ alone, and decomposes as a count term plus an alignment-filtered size surplus.
- (vi) Three exact reformulations (§10): a faithful integer conjugate T (Theorem 3) built on the identity that Step 2 is the doubled alternating bit-sum (Lemma 12); a natural-scale gap-vector conjugate (Proposition 6); and the scalar-counter form (Proposition 7).
- (vii) An extended, independently cross-validated computation. The record reaches $s_{26} = 88\,613\,778$, corroborated by $\nu(L_{26}) = 222\,569\,758$; two-step monotonicity $s_n \geq s_{n-2}$ holds for $3 \leq n \leq 26$; a run of six consecutive even-parity steps occurs across which the counter nevertheless grows; the open obstruction is localized to even steps with odd-parity predecessors (§12.1).
- (viii) A discussion (§13) of why the problem resists elementary methods, placing the forward and backward routes side by side.

2. ELEMENTARY STRUCTURE AND THE WORKED TRAJECTORY

Running the rule once by hand fixes ideas. From $L_0 = 10$ the single 1 makes $k = 1$ odd, so Step 1 appends 0001 and sets $s = -1$, giving $L^{(1)} = 100001$. The 1's lie at positions 1 and 6; filling positions 1–5 and zeroing position 6 yields 111110, and the filled gap is 4, so $s = 3$. As $3 \geq 2$ we append $(0110)^1$ and read off $L_1 = 1111100110$. Continuing,

$$\begin{aligned} L_2 &= 10101110111110011001100110, \\ L_3 &= 1100101101010001000100010000, \\ L_4 &= 100011011001111000011110000000011001100110011001100110, \end{aligned}$$

and the counter takes the values 3, 5, 2, 8, 31, 66, 158, ... (Table 2).

Two features organize everything. First, the sequence is *explosive*: from s_3 onward it roughly multiplies each step, and $|L_n|$ exceeds 10^8 before $n = 22$. Second, it *grazes the threshold exactly once*: $s_2 = 2$, with no margin. Any proof must survive that single near miss at $n = 2$ and then explain why it never recurs.

Lemma 1 (F strictly lengthens). *Whenever $F(L)$ is defined, $|F(L)| \geq |L| + 2$.*

Proof. Step 1 appends two or four symbols; Step 2 preserves length; Step 3 appends $4(s - 2) \geq 0$ symbols. $\square$

A first consequence, central to the backward viewpoint (§11): every string has only finitely many iterated preimages under F , since a preimage is strictly shorter; orbit membership is decidable.

Lemma 2 (Outputs end in 0). *Whenever $F(L)$ is defined, its last symbol is 0.*

Proof. Step 2 sets the position p_{2m} of the last 1 of $L^{(1)}$ to 0; no later position holds a 1, so $L^{(2)}$ ends in 0. If $s = 2$ then $F(L) = L^{(2)}$; if $s > 2$ then $F(L)$ ends in the final 0 of the last appended block 0110. $\square$

3. ODD PARITY FORCES SURVIVAL

Lemma 3 (Odd-parity lemma). *If $\nu(L)$ is odd, then the counter computed during $F(L)$ satisfies $s \geq 2$.*

Proof. Write $k = \nu(L)$ odd, $b \geq 0$ the number of trailing 0's of L , so $L = L_f \cdot 1 \cdot 0^b$. Step 1 appends 0001, producing $L^{(1)} = L_f \cdot 1 \cdot 0^{b+3} \cdot 1$. Now $\nu(L^{(1)}) = k + 1$ is even, and the final two 1's of $L^{(1)}$ are separated by exactly $b + 3$ zeros, so the last pair contributes $b + 3 \geq 3$. Every pair contributes ≥ 0 and s began at -1 , whence $s \geq -1 + (b + 3) \geq 2$. $\square$

From here on only even-parity strings can halt. With s starting at 0,

$$s = \sum_{j=1}^m (p_{2j} - p_{2j-1} - 1) \quad (\nu(L) \text{ even}), \quad (1)$$

a sum of nonnegative within-pair gaps. Nothing forbids $s \in \{0, 1\}$; locating those occurrences is the next task.

4. THE RUN-LENGTH CALCULUS

Write the run-length encoding of L (with $\nu(L) \geq 1$) as

$$L = 0^{c_0} 1^{r_1} 0^{c_1} 1^{r_2} 0^{c_2} \dots 1^{r_R} 0^{c_R}, \quad (2)$$

with $R \geq 1$, each $r_i \geq 1$, each internal $c_i \geq 1$. Put $C_i = r_1 + \dots + r_i$, so $C_0 = 0$ and $C_R = \nu(L)$.

Proposition 1 (Counter formula). *If $\nu(L)$ is even, the counter computed during $F(L)$ is*

$$s = \sum_{\substack{1 \leq i \leq R-1 \\ C_i \text{ odd}}} c_i. \quad (3)$$

Proof. Step 1 adds no 1's, so (1) applies. Number the 1's $1, \dots, N$; Step 2 pairs them $\{1, 2\}, \{3, 4\}, \dots$, and each pair contributes the number of 0's between its two 1's. The boundary between run i and run $i + 1$ is spanned by a pair iff C_i is odd, contributing c_i ; same-run pairs contribute 0. Summing gives (3). $\square$

Corollary 1 (Halting dichotomy). *Let $\nu(L)$ be even. Then $F(L)$ halts iff exactly one of: (i) every run length is even ($s = 0$, equivalently $\omega(L) = 0$); or (ii) the run lengths contain exactly two odd values at consecutive positions $j_0, j_0 + 1$, with $c_{j_0} = 1$ ($s = 1$).*

Proof. By (3) and $c_i \geq 1$ internally, s counts internal indices with C_i odd, weighted by c_i . The parity sequence $C_0, \dots, C_R$ starts and ends even and toggles at odd-length runs; the two stated profiles are exactly $s = 0$ and $s = 1$. $\square$

Combined with Lemma 3, the dichotomy reduces Theorem 1 to a statement about shape.

5. STRUCTURAL LEMMAS

Fix L , let $\Lambda = L \cdot (00 \text{ or } 0001)$, and let the consecutive pairs of 1's in Λ have within-pair gaps $g_j \geq 0$ for $j = 1, \dots, m = \frac{1}{2}\nu(\Lambda)$. Write C_i, γ_i for the cumulative 1-counts and internal 0-runs of Λ , and set $X(L) = \#\{i : C_i \text{ odd and } \gamma_i \text{ odd}\}$.

Lemma 4 (Reshaping). *The image $L^{(2)}$ has exactly m runs of 1's, the j -th of length $g_j + 1$; hence $F(L)$ has 1-runs $g_1 + 1, \dots, g_m + 1$ followed by $s - 2$ runs of length 2. A body run is odd iff its pair has even gap; the appended tail contributes only even-length runs.*

Proof. Pair j sets $p_{2j-1}, \dots, p_{2j} - 1$ to 1 (a block of length $g_j + 1$) and zeros p_{2j} ; distinct blocks are separated by the zeroed position. Each factor 0110 contributes one run 11 of length 2. Finally $g_j + 1$ is odd iff g_j is even. $\square$

Lemma 5 (Reshaping count). *For every L , $\omega(F(L)) = m - X(L)$.*

Lemma 6 (Odd-run floor from intra-run pairs). *With μ the number of pairs both of whose 1's lie in a single 1-run of Λ and R' the number of 1-runs of Λ , $F(L)$ has at least μ odd-length 1-runs, and $\mu \geq m - (R' - 1)$.*

Lemma 7 (The tail manufactures odd runs). *If $L = W \cdot (0110)^u$ with $u \geq 1$ and W ending in 0, then $\omega(F(L)) \geq u - \varepsilon$, where $\varepsilon = 0$ if $\nu(L)$ even and $\varepsilon = 1$ if $\nu(L)$ odd.*

Lemma 8 (Counter floor). *If $\nu(L)$ is even, then $s(F(L)) \geq \lceil \omega(L)/2 \rceil$.*

Lemma 9 (Growth). *If $L = W \cdot (0110)^u$ with $u \geq 1$, W ending in 0, and $\nu(L)$ odd, then $s(F(L)) \geq 2u + 2$.*

Proofs of Lemmas 5–9 are the pairing analysis of §4 applied run by run; they are routine and we omit them, having verified each against the engines of Appendix A.

6. PREFIX DYNAMICS AND THE ALL-EVEN PROFILE

We now address the first fragile profile, $s_n = 0$. By Corollary 1 this is exactly $\omega(L_n) = 0$. We exclude it unconditionally at even indices and reduce it to a weak counter bound at odd indices.

For a string L with $\nu(L) \geq 1$, let $r_1(L)$ denote the length of its first maximal 1-run, and (when L has at least two 1-runs) $c_1(L)$ the length of its first internal 0-run.

Lemma 10 (Prefix alternation). *For all $n \geq 0$,*

$$r_1(L_{n+1}) = \begin{cases} 1, & r_1(L_n) \geq 2, \\ c_1(L_n) + 1 \ (\geq 2), & r_1(L_n) = 1. \end{cases}$$

Since $r_1(L_0) = 1$, it follows that $r_1(L_n) = 1 \iff n$ even.

Proof. By Lemma 4, the first 1-run of $L_{n+1} = F(L_n)$ has length $g_1 + 1$, where g_1 is the within-pair gap of the first pair of $\Lambda_n = L_n \cdot (\text{pad})$. The pad is appended at the right, so the first 1-run of Λ_n equals $r_1(L_n)$. If $r_1(L_n) \geq 2$ the first two 1's are adjacent, $g_1 = 0$ and $r_1(L_{n+1}) = 1$. If $r_1(L_n) = 1$ and L_n has at least two 1-runs, the first pair straddles the first boundary, giving $g_1 = c_1(L_n)$ and $r_1(L_{n+1}) = c_1(L_n) + 1 \geq 2$. (If L_n has a single 1-run, then $\nu(L_n) = 1$ is odd; Step 1 appends 0001, creating a second 1-run at gap ≥ 3 , so again $r_1(L_{n+1}) \geq 2$.) With $a_n := \lceil r_1(L_n) \geq 2 \rceil$, $a_{n+1} = \neg a_n$ and $r_1(L_0) = 1$ give the closed form. $\square$

Proposition 2 (All-even profile excluded). *Recall $s_n = 0 \iff \omega(L_n) = 0$.*

- (a) *For every even n , L_n begins with a 1-run of length 1; hence $\omega(L_n) \geq 1$ and $s_n \neq 0$. This is unconditional.*
- (b) *For every odd n : if $\nu(L_n)$ is odd then $s_n \geq 2$ by Lemma 3; if $\nu(L_n)$ is even and $s_{n-2} \geq 4$, then $\omega(L_n) \geq 1$, so $s_n \neq 0$.*

Proof. (a) By Lemma 10, $r_1(L_n) = 1$ for even n , so $\omega(L_n) \geq 1$ and $s_n \neq 0$ by Corollary 1. (b) The odd- ν sub-case is Lemma 3. For the even- ν sub-case, $L_{n-1} = L_{n-2}^{(2)} \cdot (0110)^u$ with $u = s_{n-2} - 2$ and $L_{n-2}^{(2)}$ ending in 0. If $s_{n-2} \geq 4$ then $u \geq 2$, and Lemma 7 gives $\omega(L_n) \geq u - 1 \geq 1$. $\square$

Remark 1. The asymmetry is genuine: at an odd index with $\nu(L_n)$ even, the leading run can be even (Lemma 10 gives $r_1 = c_1 + 1$, of unconstrained parity), so part (a)'s direct argument does not transfer, and one falls back on the tail count of part (b). On the actual trajectory $s_{n-2} \geq 4$ holds at every relevant step (Table 2); the only index with $s_{n-2} < 4$ is $n - 2 = 2$ (where $s_2 = 2$), and $n = 4$

has $\nu(L_4)$ odd, so the hypothesis of part (b) is never invoked at a vulnerable step. Thus on the trajectory the all-even profile is excluded throughout.

Remark 2 (The all-even profile is reachable as an output). One might hope to prove the impossibility of $s_n = 0$ through the backward range analysis, by arguing that every output of F carries a 0 strictly between its first and last 1, so that no iterate is a single 1-run $1^a 0^b$ and the simplest all-even shape would be unreachable. *This is not so.* The odd pad 0001 places the (subsequently zeroed) last 1 at the very end of $L^{(1)}$, leaving no internal 0 in $L^{(2)}$; explicitly,

$$F(1) = 11110 = 1^4 0 \quad (\text{an output of } F, \omega = 0), \quad F(11110) = \text{halt with } s = 0.$$

Thus the all-even profile *is* reachable as an output of F , and the exclusion of $s = 0$ is a property of particular orbits, not of the range of F . The exclusion on the orbit of 10 rests on Proposition 2, which uses only the prefix invariant (Lemma 10), together with the finite backward decision procedure of §11, which confirms independently that all 4565 halting strings of length ≤ 14 are absent from the orbit of 10; neither establishes unreachability of all-even profiles in general. (A second, formally independent certificate of the orbit-of-10 all-even exclusion is given in the companion seed-independent treatment by encoding the process as a 2-state, 5-symbol Turing machine and citing a Finite Automata Reduction non-halting certificate from the Busy Beaver Challenge; we do not reproduce it here.)

For the remainder, “halting” may be read as “ $s_n = 1$.”

Proposition 3 (Reduction to a single profile). *The trajectory halts iff some $s_n = 1$.*

Proof. Halting at step n means $s_n \in \{0, 1\}$; Proposition 2 excludes $s_n = 0$ on the trajectory. $\square$

7. THE REDUCTION TO A SCALAR INEQUALITY

For $n \geq 1$ the iterate has the form $L_{n-1} = L_{n-1}^{(2)} \cdot (0110)^u$ with $u = s_{n-2} - 2$ and $L_{n-1}^{(2)}$ ending in 0 (Lemma 4).

Proposition 4 (Reduction). *If the process halts and N is the first index at which it does, then $s_{N-2} \leq 5$.*

Proof. Since N is the first halt, $s_{N-2} \geq 2$, so $u = s_{N-2} - 2 \geq 0$. If $u = 0$ then $s_{N-2} = 2 \leq 5$. If $u \geq 1$, apply Lemma 7 to $L_{N-1} = L_{N-1}^{(2)} \cdot (0110)^u$: $\omega(L_N) \geq u - 1 = s_{N-2} - 3$. By Proposition 3 the halt is $s_N = 1$, so $\omega(L_N) = 2$; hence $s_{N-2} - 3 \leq 2$. $\square$

Corollary 2 (The remaining content of the theorem). *The process never halts provided $s_n \geq 6$ for all $n \geq 3$; a fortiori it never halts if the two-step monotonicity $s_n \geq s_{n-2}$ holds for all $n \geq 3$.*

Proof. If first halt at N , Proposition 4 gives $N - 2 \leq 2$, i.e. $N \leq 4$; but $s_0, \dots, s_4 = 3, 5, 2, 8, 31$ are all ≥ 2 . Monotonicity propagates s_0, s_1, s_2 on the two parity subsequences. $\square$

Thus Theorem 1 is equivalent to the scalar statement that the counter never returns to $\{2, 3, 4, 5\}$ after the third step, for which two-step monotonicity is sufficient. The mechanism is explicit. At an odd-parity step, Lemma 9 gives

$$s_n \geq 2s_{n-1} - 2 \quad (\nu(L_n) \text{ odd}), \quad (4)$$

so odd steps at least double the counter. At an even-parity step, Lemmas 8 and 7 give

$$s_n \geq \lceil \omega(L_n)/2 \rceil \geq \lceil (s_{n-2} - 3)/2 \rceil \quad (\nu(L_n) \text{ even}), \quad (5)$$

at most a near-halving. Estimate (5) alone does not yield $s_n \geq s_{n-2}$; a long run of consecutive even-parity steps could compound several near-halvings. The next conjecture asserts this never happens.

Conjecture 1 (Two-step monotonicity; open). For all $n \geq 3$, $s_n \geq s_{n-2}$.

7.1. An improved even-step floor and a proven sub-case. We beat the factor $\frac{1}{2}$ of (5) in a structurally identified case. The key is that when the predecessor step is of even parity, the tail it carries is reshaped into a regular comb of single 1's separated by 0-gaps of length 3.

Theorem 2 (Even-step floor with even-parity predecessor). *Suppose step n is of even parity, step $n - 1$ is also of even parity, and $u := s_{n-2} - 2 \geq 1$. Then the last u body 1-runs of L_n all have length 1 and are separated by internal 0-runs of length exactly 3, and consequently*

$$s_n \geq 3 \left\lfloor \frac{u-1}{2} \right\rfloor \geq \frac{3}{2}s_{n-2} - 6. \quad (6)$$

In particular $s_n \geq s_{n-2}$ whenever $s_{n-2} \geq 12$, so Conjecture 1 holds at every even-parity step whose predecessor is of even parity, with finitely many small cases to check directly.

Proof. Write $L_{n-1} = Y \cdot (0110)^u$ with $Y = L_{n-2}^{(2)}$ ending in 0, $u = s_{n-2} - 2$. The tail carries $2u$ ones (even), and $\nu(L_{n-1})$ is even, so $\nu(Y)$ is even. Forming L_n : Step 1 appends 00, the $\nu(Y)$ ones of Y pair among themselves, and the $2u$ tail ones pair within the tail as $(1, 2), (3, 4), \dots$

The tail attached to Y reads $0 \underbrace{11001100 \cdots 0011}_u 0$. By Lemma 4 each block becomes a 1-run of length 1, Step 2 zeros its right one, and between two consecutive blocks the symbols are the zeroed right one plus the separating 00—three 0's. So the image of the tail region is $(1000)^{u-1} 1$: u single-1 runs separated by $u - 1$ internal 0-runs of length 3, lying strictly inside L_n .

Across these u runs the cumulative count C rises by 1 per run, so the $u - 1$ separators sit at $u - 1$ consecutive values of C , at least $\lfloor (u - 1)/2 \rfloor$ of them odd, each contributing $c_i = 3$ via (3). Hence $s_n \geq 3 \lfloor (u - 1)/2 \rfloor \geq \frac{3}{2}(u - 2) = \frac{3}{2}s_{n-2} - 6$, which is $\geq s_{n-2}$ iff $s_{n-2} \geq 12$. The only even-after-even step with $s_{n-2} < 12$ on the trajectory is $n = 3$, where $s_3 = 8 \geq s_1 = 5$ holds directly. $\square$

Remark 3. The bound (6) is consistent with the data: at every even-after-even step ($n = 3, 11, 12, 22, 23, 24, 25, 26$) the observed ratios s_n/s_{n-2} range from 1.60 to 7.33. The comb structure (last $u = s_{n-2} - 2$ body 1-runs of length 1, separators of length 3) was confirmed exactly at $n = 3, 11, 12$. The complementary case—an even step whose predecessor is of odd parity—is exactly where the tight instances of Conjecture 1 occur; see §12.1.

8. THE COUNTER-INERT TAIL AND BODY LOCALIZATION

The even-step floor of §7.1 lower-bounds s_n through $\omega(L_n)$. We now isolate a cleaner and exact statement which unifies the proved and the open cases. The principle is that the counter at an even step is a functional of the body alone: the freshly appended tail contributes nothing.

Lemma 11 (Counter-inert tail). *Let step n be of even parity, and write $L_n = B \cdot T$ with $T = (0110)^{s_{n-1}-2}$ the tail appended in forming $L_n = F(L_{n-1})$ and $B = L_{n-1}^{(2)}$ the body. Then T contributes exactly 0 to s_n , and s_n equals the within-pair-gap sum over B alone.*

Proof. Step 1 appends 00 and the 1's of L_n pair consecutively from the left. The tail carries $2(s_{n-1} - 2)$ ones, even, and $\nu(L_n)$ is even, so $\nu(B)$ is even; B -ones pair within B and tail ones pair within the tail as blocks 11 of gap 0. $\square$

Combining Lemma 11 with (3) yields an exact decomposition of the even-step counter.

Observation 1 (Body localization and the count/surplus split). At an even-parity step n , the nonzero within-pair gaps occupy exactly the first $\nu(L_{n-1}^{(2)})/2$ pair-slots—the body—and the trailing pairs sum to 0. Each contributing body pair carries a within-pair gap $g \geq 1$; writing $g = 1 + (g - 1)$ partitions the counter exactly as

$$s_n = \underbrace{\#\{\text{body boundaries with odd cumulative 1-count}\}}_{\text{count term}} + \underbrace{\sum_{\text{those boundaries}} (g - 1)}_{\text{size surplus}}. \quad (7)$$

The count term is bounded below by the floor $\lceil \omega(L_n)/2 \rceil$ of Lemma 8, with equality only when each parity-raised interval of the cumulative-1-count contributes a single boundary—an extremal case that does not occur at tight instances of Conjecture 1. The size surplus is carried by the contributing within-pair gaps of size ≥ 2 .

The decomposition (7) was confirmed quantitatively at every odd-predecessor even step within reach (Appendix B, Table 1). In each case the freshly appended tail contributes 0, in exact agreement with Lemma 11. At the binding case $n = 15$ the contributing gaps are dominated by 1's—the signature of the cross-paired comb produced by an odd-parity predecessor—the count term is 23,273 (strictly exceeding the lower bound $\lceil \omega(L_{15})/2 \rceil = 22,455$), and the surplus over that count, namely $5810 = 5613 \cdot 1 + 1 \cdot 2 + 65 \cdot 3$, is supplied entirely by the gaps of size ≥ 2 , which originate, recursively, in the deeper body $L_{n-2}^{(2)}$. The true margin is $s_{15} - s_{13} = 349$.

This is the precise sense in which the residual difficulty has been relocated: by (7) the whole question is whether the size surplus is large enough; §12.1 shows that the count term, even with its strongest provable lower bound $\lceil \omega/2 \rceil$, can never reach it.

9. THE ν -RECURRENCE AND THE PARITY DYNAMICS

The parity of $\nu(L_n)$ decides which regime governs step n and is the pivotal discrete datum of the whole development. It obeys an exact recurrence.

Proposition 5 (ν -recurrence). *Whenever $F(L)$ is defined, with $\nu = \nu(L)$ and s the counter,*

$$\nu(F(L)) = \begin{cases} \frac{\nu}{2} + 3s - 4, & \nu \text{ even}, \\ \frac{\nu+1}{2} + 3s - 3, & \nu \text{ odd}. \end{cases}$$

Consequently the parity of $\nu(L_{n+1})$ is determined by $(\nu(L_n) \bmod 4, s_n \bmod 2)$; for even parity, $\nu(L_{n+1}) \equiv \nu(L_n)/2 + s_n \pmod{2}$.

Proof. $\nu(F(L)) = \nu(L^{(2)}) + 2(s - 2)$. By Lemma 4, $L^{(2)}$ has m runs of total 1-length $m + \sum_j g_j$. For even parity, $\sum_j g_j = s$ and $m = \nu/2$, so $\nu(L^{(2)}) = \nu/2 + s$; for odd parity, $\sum_j g_j = s + 1$ and $m = (\nu + 1)/2$, so $\nu(L^{(2)}) = (\nu + 1)/2 + s + 1$. Substitute. $\square$

Iterating Proposition 5 from $\nu(L_0) = 1$ with the computed counters reproduces the observed parity string exactly (Appendix B), and yields the parity at $n = 26$ without materializing L_{26} . The counter s_n itself admits no scalar recurrence: by Proposition 1 it depends on the full pattern of odd-cumulative gaps of L_n .

10. EQUIVALENT FORMULATIONS

We record three reformulations of the process. Throughout, $w(x)$ denotes the popcount of x and $V(L) := \text{int}(L, 2)$ the binary value.

10.1. The pair-and-fill identity. For an integer M with $w(M) = 2m$ and 1-bit positions $b_1 > b_2 > \dots > b_{2m} \geq 0$, set

$$S(M) := \sum_{i=1}^{2m} (-1)^{i+1} 2^{b_i}.$$

Lemma 12 (Step 2 is doubling of the alternating bit-sum). *If $\nu(L)$ is even and $M = V(L^{(1)})$, then $V(L^{(2)}) = 2S(M)$. The pair (b_{2j-1}, b_{2j}) effects the local change $2^{b_{2j-1}} - 3 \cdot 2^{b_{2j}}$ to M .*

Proof. The left/right members of pair j sit at heights $A_j = b_{2j-1} > B_j = b_{2j}$. Step 2 replaces $2^{A_j} + 2^{B_j}$ by $\sum_{i=B_j+1}^{A_j} 2^i = 2^{A_j+1} - 2^{B_j+1}$; the local change is $2^{A_j} - 3 \cdot 2^{B_j}$. Summing, $V(L^{(2)}) = \sum_j (2^{A_j+1} - 2^{B_j+1}) = 2 \sum_i (-1)^{i+1} 2^{b_i} = 2S(M)$. $\square$

The constant 3 here is the genuine arithmetic kernel of the process, the structural counterpart of the 3 in the Collatz map.

10.2. The integer form. By Lemma 10 the leading 0-run of every iterate is empty, so each L_n begins with 1; each ends in 0 (Lemma 2). Hence V is a bijection from the reachable strings onto $\mathbb{Z}^+$, with $V(L_0) = 2$.

Theorem 3 (Integer conjugate). *Through V , F is conjugate to $T : \mathbb{Z}^+ \dashrightarrow \mathbb{Z}^+$ given by*

$$T(N) = 2S(M) \cdot 16^{s-2} + 6 \cdot \frac{16^{s-2}-1}{15}, \quad M = \begin{cases} 4N, & w(N) \text{ even}, \\ 16N + 1, & w(N) \text{ odd}, \end{cases}$$

$$s = s_{\text{init}} + w(S(M)) - \frac{1}{2} w(M), \quad s_{\text{init}} = \begin{cases} 0, & w(N) \text{ even}, \\ -1, & w(N) \text{ odd}, \end{cases}$$

defined exactly when $s \geq 2$. In hexadecimal, $T(N)$ is the digit string of $2S(M)$ followed by $s - 2$ copies of the digit 6. The seed is $N_0 = 2$.

Proof. Step 1 multiplies by 4 (pad 00) or 16 then +1 (pad 0001), giving M . Step 2 gives $V(L^{(2)}) = 2S(M)$ (Lemma 12). The counter is $s = s_{\text{init}} + \nu(L^{(2)}) - m = s_{\text{init}} + w(S(M)) - \frac{1}{2} w(M)$. Step 3, when $s \geq 2$, appends $(0110)^{s-2}$; since $0110_2 = 6$ fills one hex digit, this is multiplication by 16^{s-2} followed by adding $\underbrace{6 \cdots 6}_{s-2} = 6(16^{s-2} - 1)/15$. $\square$

Remark 4 (How close to Collatz?). The shape is Collatz-like: a parity test on $w(N)$ selects one of two affine pads, the kernel carries the constant 3 (Lemma 12), and the whole question reduces to one inequality on a derived scalar. Two features are not Collatz-like, and both are intrinsic. First, the orbit grows doubly exponentially,

$$N_0, N_1, N_2, N_3, \dots = 2, 998, 45\,868\,646, 213\,192\,976, \dots,$$

because the binary value carries $\Theta(|L_n|)$ digits and $|L_n| \sim 4^{s_n}$ grows geometrically. Second, S is non-local: it depends on the global ordering of all set bits rather than on N through a fixed affine rule.

10.3. The gap-vector form. Encode a string that begins with 1 and ends in 0 by the gaps between its consecutive 1's together with its trailing-zero count,

$$(e; t), \quad e = (e_1, \dots, e_{N-1}) \in \mathbb{Z}_{\geq 0}^{N-1}, \quad t \geq 1,$$

where $N = \nu(L)$. The seed is $((); 1)$.

Proposition 6 (Gap-vector conjugate). *Write $N = \nu$. One step of F is:*

- (pad; seed the counter) *If N is even, set $s_{\text{init}} := 0$ and leave $(e; t)$ unchanged. If N is odd, set $s_{\text{init}} := -1$, append $e_N := t + 3$, reset $t := 0$, and increment N . Now N is even.*
- (counter) $s = s_{\text{init}} + (e_1 + e_3 + e_5 + \dots + e_{N-1})$; *halt if $s < 2$.*
- (rewrite) *The new word of 1-runs and separators is*

$$1^{e_1+1} 0^{1+e_2} 1^{e_3+1} 0^{1+e_4} \dots 1^{e_{N-1}+1} \text{ followed by } (0110)^{s-2}.$$

The new trailing-zero count is $t' = 1$ when $s > 2$, and $t' = t + 3$ when $s = 2$.

Proof. This is the run-length calculus of §4 in gap coordinates. Pair j has within-pair gap e_{2j-1} and yields a block of $e_{2j-1} + 1$ ones; its zeroed right endpoint together with the between-pair gap e_{2j} makes the separator $1 + e_{2j}$. The counter is $s_{\text{init}} + \sum_j e_{2j-1}$. The odd- N case is the 0001 pad. When $s > 2$ the appended factor ends in 0, $t' = 1$; when $s = 2$ the word ends at the zeroed right endpoint of the last pair, after which lie the $t + 2$ trailing zeros of the padded string (two from the 00 pad), giving $t' = t + 3$. $\square$

Remark 5 (Two subtleties of the encoding). Two points are easy to get wrong and worth isolating. First, the additive seed $s_{\text{init}} = -1$ of the odd-parity pad must be carried in the counter line; without it the seed step returns $s = 4$ in place of $s_0 = 3$. Second, the trailing-zero count is not always 1: the unique $s = 2$ step ($n = 2$) produces L_3 with four trailing zeros ($t = 1$, so $t' = t + 3 = 4$), so t must be tracked honestly. With both handled as stated, the gap-vector engine reproduces the entire trajectory exactly.

10.4. The scalar form. Collapsing to the counter alone: by Proposition 2 the all-even profile never occurs on the trajectory, and then (Proposition 3) a halt is exactly some $s_n = 1$. With Corollary 2 this gives the following.

Proposition 7 (Scalar form). *On the trajectory, F is non-halting if and only if the integer sequence $(s_n)_{n \geq 0}$ never re-enters $\{2, 3, 4, 5\}$ after $n = 2$; a clean sufficient condition is two-step monotonicity $s_n \geq s_{n-2}$ for $n \geq 3$ (Conjecture 1). The sequence satisfies $s_n \geq 2s_{n-1} - 2$ at popcount-odd steps and $s_n \geq \lceil (s_{n-2} - 3)/2 \rceil$ at popcount-even steps, but admits no closed recurrence.*

This last point is the precise sense in which the problem is of Collatz type, and it is the wall on which both the forward inequality and the backward search (§11) rest.

11. THE BACKWARD VIEWPOINT

A complementary strategy reasons from a hypothetical halt. Its foundation is the finiteness noted after Lemma 1: preimages are strictly shorter, so the backward tree of any string is finite and orbit membership is decidable. In fact F is exactly invertible.

Proposition 8 (Invertibility). *Let $T \in \{0, 1\}^*$. Every X with $F(X) = T$ arises as follows: choose $u \geq 0$ with $T = Y \cdot (0110)^u$ and Y ending in 0; from the maximal 1-blocks $[a, b]$ of Y form the candidate $L^{(1)}$ with 1's exactly at $\{a, b + 1\}$ over the length of Y ; delete a suffix 00 or 0001 to obtain X ; and retain X iff $F(X) = T$. There are finitely many preimages.*

Proof. If $F(X) = T$ then $T = L^{(2)} \cdot (0110)^{s-2}$ with $L^{(2)}$ ending in 0, so T has the stated form with $u = s - 2$ and $Y = L^{(2)}$. Step 2 turns each pair (p_{2j-1}, p_{2j}) into the block $[p_{2j-1}, p_{2j} - 1]$ and zeros p_{2j} , so the blocks $[a, b]$ of Y recover the pair endpoints, determining $L^{(1)}$ uniquely. Verifying $F(X) = T$ discards spurious reconstructions. Finiteness is Lemma 1. $\square$

Proposition 8 turns the backward search into a finite, exact computation. Suppose the process first halts at index $N \geq 2$. By Proposition 4, $s_{N-2} \leq 5$. We now ask what is forced about the predecessor counter s_{N-1} and the shape of L_N .

Proposition 9 (Shape of a first halt). *If the process first halts at $N \geq 2$, then $s_{N-2} \leq 5$ (Proposition 4), and by Lemma 11 the halt depends only on the body $B = L_{N-1}^{(2)}$: the tail $(0110)^{s_{N-1}-2}$ of L_N contributes 0 to s_N . Consequently the predecessor counter s_{N-1} is not determined by the halt and may be arbitrarily large.*

Proof. Proposition 4 gives $s_{N-2} \leq 5$. At the halt $\nu(L_N)$ is even, so Lemma 11 applies to the step computing s_N , which therefore depends on B alone, independently of the tail length $s_{N-1} - 2$. $\square$

Remark 6 (The predecessor counter is not pinned at a first halt). It is tempting to think that a first halt forces $s_{N-1} = 2$, so that the halting string has an empty tail $L_N = L_{N-1}^{(2)}$ and consists of m image blocks summing to $m + 2$ ones. *This is not so:* the pinning $s_{N-1} = 2$ would contradict Lemma 11, which makes the halt independent of s_{N-1} . Two explicit counterexamples:

- the seed 1100111 first halts at $N = 2$ with $s_{N-1} = 3$, $s_N = 1$;
- the seed 110011111110 first halts at $N = 8$ with $s_{N-1} = 102$, $s_N = 1$.

The subtlety is that the tail's 11 blocks are even runs of L_N , not odd runs, so Lemma 8 yields only $\lceil \omega/2 \rceil = 1$ rather than $s_N \geq 2$. The halting strings are therefore $B \cdot (0110)^k$ ($k \geq 0$) with B fragile.

The finite backward decision procedure is exact regardless. By Lemma 1 the iterated preimages of any fixed string form a finite tree, so “does w ever occur on the trajectory?” is decided by tracing that tree to its roots and testing membership of $L_0 = 10$. Running this decision procedure wholesale, every halting string of length at most 14—there are 4565 of them, of which 3137 have three or more 1-runs—is found to be absent from the trajectory. The procedure does not extend to a closed-form argument for unbounded lengths: backward chains lengthen with size, so the backward route demands a uniform descent of strength equal to the two-step monotonicity $s_n \geq s_{n-2}$. As §13 makes precise, this is the same wall the forward route meets.

12. COMPUTATIONAL RESULTS

All computations were carried out by two independent implementations of Definition 1, described in Appendix A: a direct bit-level simulator and a run-length engine acting on the gap sequence of the 1’s. The two agree on every output where both are feasible—full strings through $n = 12$, and the scalar quantities s_n and $\nu(L_n) \bmod 2$ through $n = 25$ —and the run-length engine was further cross-checked against the ν -recurrence (Proposition 5). Every figure reported here was reproduced by at least two independent computations; the external verification is described in Appendix D. Counter and parity. Table 2 lists s_n and the parity of $\nu(L_n)$ for $0 \leq n \leq 26$. Every $s_n \geq 2$; the unique near miss is $s_2 = 2$.

Two-step monotonicity. The inequality $s_n \geq s_{n-2}$ holds for every $3 \leq n \leq 26$. The tightest instance is the odd $\rightarrow$ even transition $n = 15$, where $s_{15}/s_{13} = 29083/28734 \approx 1.0121$.

A long even-parity run. The parity of $\nu(L_n)$ is even at $n = 2, 3, 10, 11, 12, 15, 19, 21, 22, 23, 24, 25, 26$. In particular $n = 21, \dots, 26$ are six consecutive even-parity steps—precisely the configuration whose danger (5) cannot a priori rule out. Yet across this run the counter strictly increases at every step,

$$s_{21..26} = 3\,732\,091, 8\,693\,523, 18\,136\,691, 24\,703\,847, 55\,831\,137, 88\,613\,778,$$

so the feared compounding of near-halvings does not occur. As $n = 26$ is even-after-even, it falls in the regime already settled by Theorem 2.

12.1. The residual obstruction, localized. The even-parity steps split by the parity of their predecessor:

predecessor even (Theorem 2 applies)	$n = 3, 11, 12, 22, 23, 24, 25, 26$; ratios 1.60–7.33
predecessor odd (open)	$n = 10, 15, 19, 21$; ratios 1.36, 1.01, 1.32, 5.66

Every tight instance of Conjecture 1— $n = 15$ (1.012), $n = 19$ (1.324), $n = 10$ (1.361)—has an odd-parity predecessor. The structural reason is that an odd-parity predecessor forces cross-pairing of its tail (Lemma 7, $\varepsilon = 1$), producing odd runs separated by small gaps rather than the size-3 comb of Theorem 2; at $n = 15$ the counted gaps average 1.25, so the floor collapses to the factor $\frac{1}{2}$ of (5).

By the decomposition (7) of Observation 1, the entire residual difficulty is the size-surplus term. At the binding case the strongest provable lower bound on the count term—the floor $\lceil \omega(L_n)/2 \rceil$ of Lemma 8—is already insufficient: directly,

$$\lceil \omega(L_{15})/2 \rceil = 22\,455 < s_{13} = 28\,734, \quad \omega(L_{15}) = 44\,910 < 2\,s_{13},$$

so any argument of the form “ $s_n \geq \frac{1}{2}\omega(L_n)$ with $\omega(L_n) \geq 2s_{n-2}$ ” is dead on arrival. Concretely, the contributing-gap multiset at $n = 15$ is $\{1 : 17\,594, 2 : 5613, 3 : 1, 4 : 65\}$ (Table 1): the count term contributes the 23 273 boundaries—strictly exceeding the lower bound 22 455—and the surplus over that count, namely $5613 \cdot 1 + 1 \cdot 2 + 65 \cdot 3 = 5810$, is supplied entirely by the gaps of size ≥ 2 , which originate, recursively, in the deeper body $L_{n-2}^{(2)}$. The true margin is $s_{15} - s_{13} = 349$.

This is the entire residual content of Conjecture 1:

prove $s_n \geq s_{n-2}$ at an even-parity step n whose predecessor is of odd parity,

a statement requiring control of the alignment-filtered sizes of the cross-paired gaps, not merely of odd-run counts. The surplus is a sum of the predecessor’s within-pair gaps restricted to the boundaries at which the running 1-count is odd; that restriction is an alignment depending on the parities of all earlier run lengths, hence on the full history—by Proposition 7, on a quantity with no scalar recurrence. No descent or Lyapunov function in the computable data (s_n , $s_n + s_{n-1}$, ratios, ω , gap-moment sums) was found that is preserved across the $s_2 = 2$ near miss while forcing $s_n \geq 6$ thereafter. The inequality thus appears to lie beyond elementary methods; §13 develops why, and §14 states the status precisely.

13. WHY THE PROBLEM RESISTS

It is worth setting out, concretely, why a problem whose answer is in so little doubt should be so hard to prove. The obstruction is not a shortage of structure—the process is unusually transparent—but a specific mismatch between the quantity one must control and the quantities one can compute. We organise the discussion around the two natural proof strategies, which turn out to fail at the same point.

13.1. Two routes, one obstruction. The forward route sweeps along the trajectory and asks how a single application of F reshapes runs. By the reshaping identities (Lemmas 4–6) the odd-length runs of L_{n+1} are exactly the images of the even-gap pairs of L_n , the appended tail contributes only even runs, and intra-run pairing manufactures odd runs in profusion. To forbid both fragile profiles at once it suffices to know that every even-parity iterate carries at least four odd-length runs (four, not two, because the count is necessarily even). Call this *Invariant (A)*. It instantly excludes $\omega = 0$ and $\omega = 2$, and with Lemma 3 it would finish Theorem 1.

The difficulty is that the only uniform lower bound available, $\mu \geq m - (R' - 1)$ of Lemma 6, collapses precisely when the string degenerates to a near-comb of length-one runs: for $L = 1010 \cdots 10$ one has $\nu(\Lambda) \approx R'$ and the bound gives no floor at all. Invariant (A) therefore cannot be read off a single step; it is a statement about the orbit never degenerating to such a comb at an even step.

The backward route assumes a first halt at N and analyses L_N . By Proposition 9, $s_{N-2} \leq 5$ and the halt is a functional of the body $B = L_{N-1}^{(2)}$ alone—the tail of L_N is counter-inert. The analysis of Remark 6 allows nonempty tails: the halting strings are $B \cdot (0110)^k$ with B fragile, and the finite preimage decision procedure (Proposition 8) traces the backward tree of any B to its roots and tests membership of L_0 . This procedure dispatches every halting string of length ≤ 14 , but it does not extend to a closed-form argument: backward chains lengthen with $|B|$, and what is required to rule out fragile B uniformly is precisely Invariant (A)—that no L_n ever lands in a fragile shape. The forward route calls it Invariant (A); the backward route needs it to retire the unbounded family of fragile bodies B . The two decisive steps coincide.

13.2. Why that obstruction is a wall. Three features, each intrinsic, conspire to keep Invariant (A) out of reach.

The counter is a parity functional, small only on fragile shapes. By Proposition 1, s adds up the internal gaps that sit after an odd cumulative count of 1’s. This is a global parity selection, not a local or affine readout: which gaps are counted depends on the parities of all preceding run lengths. A parity functional can be small only on configurations of a very particular shape, so the entire content of the theorem is that the dynamics steer the orbit clear of those shapes—and no amount of cleverness about a single step controls a global parity alignment.

The decisive quantity is a size surplus, not a count. The exact decomposition (7) splits the even-step counter into a count term, bounded below by the floor $\lceil \omega/2 \rceil$, and a size surplus carried by the within-pair gaps of size ≥ 2 at odd boundaries. At the binding case $n = 15$ the count floor $\lceil \omega(L_{15})/2 \rceil = 22\,455$ is already smaller than $s_{13} = 28\,734$, so no argument bounding odd-run counts from below by $\lceil \omega/2 \rceil$ can close the gap (§12.1); the whole margin comes from the sizes of a few gaps,

and those sizes are themselves alignment-filtered, depending on the full history. One is asked to control not how many odd runs there are, but how large the right-aligned ones are—a strictly finer datum.

There is no scalar recurrence, and the single near miss forbids crude monotonicity. By Proposition 7 the counter obeys no closed recurrence: s_{n+1} depends on the entire odd-cumulative gap pattern of L_n , not on finitely many earlier counters. This is the precise sense in which the system is of Collatz type, and through the integer conjugate T (Theorem 3) the kinship is exact—a popcount-parity test selecting one of two affine pads, with the kernel carrying the Collatz constant 3 (Lemma 12). Worse for any monotone approach, the trajectory touches the threshold with equality exactly once, at $s_2 = 2$. A successful argument must therefore permit the counter to descend all the way to 2 once and then explain—structurally—why it never descends again.

Growth makes finite verification powerless as a substitute. The integer orbit grows doubly exponentially (Remark 4), because the binary value carries $\Theta(|L_n|)$ digits and $|L_n| \sim 4^{s_n}$ grows geometrically; the run count itself roughly triples per step. No finite computation, however far pushed, can stand in for the missing uniform argument.

13.3. What the evidence does and does not settle. The reductions are complete and rigorous: Theorem 1 is equivalent, modulo the proved Lemmas 3, 4 and Proposition 1, to the single statement that no even-parity iterate wears a fragile profile, for which two-step monotonicity is a clean sufficient condition. That statement holds for every iterate within computational reach—the even members carry 4, 6, 1420, 3704, 1960, 44910, ... odd runs, never the 0 or 2 a halt would require—and for every halting string of length at most 14 by the backward decision procedure. The counter’s geometric recovery after its lone near miss, the strict growth even across six consecutive even-parity steps, two independent simulators in agreement, and the external verification of Appendix D all point the same way. What remains is a single, sharply isolated inequality whose proof would require a size-aware estimate of the cross-paired gap structure that the present methods do not supply.

14. STATUS

Proved. F strictly lengthens and outputs end in 0 (Lemmas 1, 2); odd parity forces $s \geq 2$ (Lemma 3); for even parity the counter is the odd-cumulative gap sum (3), whence the halting dichotomy (Corollary 1); the reshaping identities (Lemmas 4–9); the prefix-alternation invariant $r_1(L_n) = 1 \iff n$ even (Lemma 10); the exclusion of the all-even profile, unconditionally at even indices and—via $s_{n-2} \geq 4$ —at odd indices (Proposition 2); the Reduction, that a first halt forces $s_{N-2} \leq 5$ (Proposition 4); the ν -recurrence (Proposition 5); the counter-inert tail and the exact count/surplus decomposition (Lemma 11, Observation 1); the even-step floor giving two-step monotonicity at every even step with an even-parity predecessor (Theorem 2); invertibility of F and the first-halt shape (Propositions 8, 9).

Reduced. Theorem 1 is equivalent (Corollary 2) to the scalar statement that (s_n) never re-enters $\{2, 3, 4, 5\}$ after $n = 2$, for which two-step monotonicity $s_n \geq s_{n-2}$ is sufficient. Monotonicity itself is now reduced to a single configuration: an even-parity step with an odd-parity predecessor (§12.1). The odd-parity steps ($s_n \geq 2s_{n-1} - 2$) and the even-after-even steps (Theorem 2) are settled. Within the open configuration, the difficulty is localized further by (7) onto the alignment-filtered size surplus, the count term being provably insufficient under its strongest available lower bound.

Open. The inequality $s_n \geq s_{n-2}$ at an even step with odd-parity predecessor, and hence Conjecture 1 and Theorem 1. The binding case $n = 15$ has a margin of 1.2%. The computational evidence—the counter’s geometric growth after its single near miss, the strict growth even across a run of six even-parity steps, two independent simulators in agreement, and the external verification of Appendix D—leaves little doubt that the theorem is true; what remains is the closing of one sharply delimited, structurally identified inequality.

APPENDIX A. ALGORITHMS

All computational claims rest on direct implementations of Definition 1. We summarize the two engines; both were re-implemented independently during the external verification of Appendix D, with agreement on all common outputs.

Bit-level simulator. Given L : set $k = \nu(L)$; if k is even append 00 and $s \leftarrow 0$, else append 0001 and $s \leftarrow -1$. List the 1-positions $p_1 < \dots < p_{2m}$; for $j = 1, \dots, m$ set positions $p_{2j-1}, \dots, p_{2j} - 1$ to 1, set p_{2j} to 0, and $s \leftarrow s + (p_{2j} - p_{2j-1} - 1)$. If $s < 2$ halt; otherwise return $L^{(2)} \cdot (0110)^{s-2}$. This is the literal ground truth, feasible while the string fits in memory (through $n = 12$ for full strings). Run-length / gap engine. Represent L by its run vector $[c_0, r_1, c_1, \dots, r_R, c_R]$, equivalently by the gap-vector $(e; t)$ of §10.3. One step is computed by Proposition 6: pad and seed s_{init} ; read s as the seed plus the odd-indexed gap sum; emit blocks $e_{2j-1} + 1$ and separators $1 + e_{2j}$; append the tail and update t honestly. This runs in time and space $O(\# \text{runs})$, advancing the trajectory well beyond the point where strings can be stored. The value s_{26} was obtained by a memory-frugal pass that sums separators by cumulative-count parity, never materializing the 1.7×10^8 -entry gap vector; that pass was validated against the direct value at $n = 10$ before its use at $n = 26$.

APPENDIX B. COMPUTATIONAL EVIDENCE

With $L_0 = 10$, the counter sequence $s_0, \dots, s_{26}$ is

3, 5, 2, 8, 31, 66, 158, 392, 938, 2305, 1277, 4885, 9361, 28734, 67339, 29083,
218336, 498138, 1332397, 659539, 4146749, 3732091, 8693523, 18136691,
24703847, 55831137, 88613778,

and the parity string of $\nu(L_n)$ for $0 \leq n \leq 26$ is (o = odd, E = even)

o o E E o o o o o o E E E o o E o o o E o E E E E E.

The ν -recurrence (Proposition 5), iterated from $\nu(L_0) = 1$ with these counters, reproduces this parity string exactly, and predicts $\nu(L_{26}) = 222\,569\,758$, reproduced independently by the separator-parity construction of s_{26} . Two-step monotonicity $s_n \geq s_{n-2}$ holds for all $3 \leq n \leq 26$; the tightest instance is $n = 15$ (≈ 1.0121). For the structural claim of Theorem 2, the reshaped comb was confirmed exactly at $n = 3, 11, 12$.

The odd-run counts $\omega(L_n)$ for $0 \leq n \leq 15$ are

1, 1, 4, 6, 1, 11, 47, 105, 235, 589, 1420, 3704, 1960, 7549, 13983, 44910;

the small values occur only at odd-parity iterates where survival is guaranteed by Lemma 3; at every even-parity iterate the count is well above the value 2 that a halt would require. The backward decision procedure confirms that all 4565 halting strings of length at most 14—including the 3137 with three or more 1-runs—are absent from the trajectory.

The exact body/tail decomposition of §8 was verified at the odd-predecessor even steps within reach; Table 1 records the result.

n	s_n	body Σ / tail Σ	nonzero contributing-gap multiset
10	1 277	1 277 / 0	{1 : 551, 2 : 230, 3 : 87, 5 : 1}
15	29 083	29 083 / 0	{1 : 17 594, 2 : 5613, 3 : 1, 4 : 65}
19	659 539	659 539 / 0	{1 : 273 509, 2 : 147 958, 3 : 17 266, 4 : 7769, 5 : 810, 10 : 319}

TABLE 1. Body/tail decomposition of the counter at the open (odd-predecessor) even steps. The tail Σ is identically 0 (Lemma 11). For $n = 15$, the count term is 23,273 (strictly exceeding $\lceil \omega/2 \rceil = 22,455$) and the size surplus $5613 \cdot 1 + 1 \cdot 2 + 65 \cdot 3 = 5810$, so $s_{15} = 23,273 + 5810 = 29,083$.

n	s_n	par. $\nu(L_n)$	n	s_n	par. $\nu(L_n)$
0	3	odd	14	67 339	odd
1	5	odd	15	29 083	even
2	2	even	16	218 336	odd
3	8	even	17	498 138	odd
4	31	odd	18	1 332 397	odd
5	66	odd	19	659 539	even
6	158	odd	20	4 146 749	odd
7	392	odd	21	3 732 091	even
8	938	odd	22	8 693 523	even
9	2 305	odd	23	18 136 691	even
10	1 277	even	24	24 703 847	even
11	4 885	even	25	55 831 137	even
12	9 361	even	26	88 613 778	even
13	28 734	odd			

TABLE 2. The counter s_n and parity of $\nu(L_n)$, $0 \leq n \leq 26$. Even-parity (potentially halting) steps are set in bold. The value $s_{26} = 88\,613\,778$ was obtained by a memory-frugal separator-parity pass and corroborated by the independently computed $\nu(L_{26}) = 222\,569\,758$ via Proposition 5.

APPENDIX C. COMPUTATIONAL PROGRAMS

The two implementations of Definition 1—the bit-level simulator and the run-length/gap engine of Appendix A—together with the verification and statistics-gathering scripts, are available at

<https://github.com/RobinCodes/Projects/tree/main/Collatz-type%20problem/Collatz-type%20problem%20v6>.

APPENDIX D. NOTE ON THE USE OF ARTIFICIAL INTELLIGENCE

This work was prepared with substantial assistance from an artificial-intelligence system, and most of the results presented here were created with such assistance. Machine assistance implemented the map of Definition 1 in two independent ways—the bit-level and run-length engines of Appendix A—cross-validated them on every commonly feasible output, extended the computational record of Appendix B, and helped formulate and prove the structural results: the prefix-alternation invariant (Lemma 10), the exclusion of the all-even profile (Proposition 2), the even-step floor (Theorem 2), the ν -recurrence (Proposition 5), the counter-inert tail and its exact decomposition (Lemma 11, Observation 1), the three reformulations of §10, and the localization of the residual obstruction (§12.1, §13). Computation contributed $s_{26} = 88\,613\,778$, $\nu(L_{26}) = 222\,569\,758$, and the body/tail decompositions of Table 1.

We do not specify the particular artificial-intelligence system or systems used; readers seeking further detail about the methods are invited to contact the authors: Robin Rovenszky ([robincodes](#) on Discord) and [notxxdog](#) ([.notxxdog](#) on Discord).

Every proof and every computational claim in this paper was independently verified by two human readers and by independent machine re-derivations. An independent re-derivation, working only from the statements of this paper, reconstructed the map of Definition 1 from scratch, re-implemented it in two further from-scratch engines (a literal bit-level engine and a numpy gap-vector engine), cross-validated the two engines against each other, and reproduced every computational and structural claim: the counter sequence and parity through $n = 26$ with $s_{26} = 88\,613\,778$ and $\nu(L_{26}) = 222\,569\,758$; the integer-conjugate values $N_0, \dots, N_3 = 2, 998, 45\,868\,646, 213\,192\,976$ and the conjugacy $T(N_n) = N_{n+1}$ through $n = 7$; all structural lemmas of §§4–5–6 by brute force on

strings of length ≤ 12 beginning with 1, with zero violations; the body/tail decomposition multisets of Table 1; and the count/surplus decomposition of Observation 1, with the count term identified as the number of contributing boundaries (which strictly exceeds $\lceil \omega/2 \rceil$ at tight instances). This re-derivation reached the same conclusion regarding the central inequality—that it lies beyond elementary methods.

No result here is accepted on the basis of machine assertion. Every definition, lemma, and proof is stated so as to be checkable by hand, and every computational claim is reproducible by re-running the procedures of Appendix A and the programs of Appendix C. The status of the central open inequality is reported exactly as it stands.